Impact of China's 2014 Hukou Reforms on City-Level Population Growth

Advised by Andrés Maggi

Jessica Chen

May 2025

Abstract

I study the impact of China's 2014 hukou reform on city-level population growth. Using 2010–2023 resident-population and GDP data for Zhejiang and Sichuan and a difference-in-differences design that codes reform intensity city by city, I find no average population gain in treated cities; instead, sub-provincial hubs such as Hangzhou and Chengdu keep expanding while many prefecture-level cities continue to lose residents.

1 Introduction

In this paper, I address the empirical question: *How does partial hukou liberalisation shape medium-run population growth across Chinese cities?* China's 2014 reform—officially styled the "Guiding Opinions on Further Deepening the Reform of the Household Registration System"—invited municipalities below five million inhabitants to lower settlement barriers for rural migrants. Whether this invitation succeeded in rebalancing population away from megacities toward smaller regional centres remains an open question.

Existing scholarship documents that hukou restrictions suppress labour mobility (Chan and Zhang 1999; Au and Henderson 2006) and that easing those barriers can boost innovation and firm formation (Lu and Qian 2024). However, prior work typically infers treatment status from headline population thresholds and focuses on the two years immediately following the reform. Such approaches risk both misclassification—because hukou rules are implemented at county or district level—and attenuation of long-run effects that unfold gradually. My study departs from the literature by manually parsing local regulations and news reports to code reform intensity, and by tracking outcomes through 2023.

The analysis draws on provincial statistical yearbooks, which I scraped from individual bureau websites, sometimes via Hong Kong VPN to bypass access blocks. The final panel covers Beijing, Shanghai, eleven Zhejiang, and eighteen Sichuan cities, providing annual resident-population and GDP per capita series.

I estimate a difference-in-differences specification with province, year, and city fixed effects, treating reform adoption as a binary indicator that switches on in 2015. I complement the regression with qualitative case studies—most notably the divergent experiences of Hangzhou and Wenzhou in Zhejiang, and Chengdu versus its neighbouring prefectures in Sichuan.

The baseline regression yields an imprecise estimate centred slightly below zero, indicating that, on average, hukou relaxation neither stemmed out-migration from smaller inland cities nor curtailed the inflow into established hubs. Qualitative evidence helps reconcile this null result: fiscally robust sub-provincial cities attracted migrants regardless of policy stance, whereas counties that embraced full hukou unification often lacked the labour-market depth to retain newcomers.

By combining granular policy coding with yearbook data, this paper highlights the importance of local administrative capacity in mediating national reforms. The findings caution against evaluating hukou policy through a one-size-fits-all lens and underscore the need for city-specific diagnostics when designing future household-registration initiatives.

2 Literature Review

2.1 Hukou Reform and Population Mobility

The hukou system has long constrained labor mobility in China by restricting access to urban services for migrants without local registration. Numerous studies have documented how changes to hukou policy influence migration patterns. Chan and Zhang (1999) find that hukou restrictions reduced rural-urban migration by raising the costs of urban settlement. Au and Henderson (2006) show that cities with stricter migration controls experienced slower labor market integration and lower productivity. More recent work has studied how the relaxation of hukou policies affects population growth. Liu et al. (2024) find that the 2014 hukou reform, which relaxed settlement restrictions in small and medium-sized cities, led to a significant increase in migrant inflows and new firm creation. Lu and Qian (2024) extend this analysis by documenting that cities subject to hukou relaxation also experienced increases in patenting activity. These findings suggest that easing migration barriers alters both the size and composition of urban populations, with potential consequences for local economic outcomes.

2.2 Comparison with Lu and Qian (2024)

This paper builds closely on the research design of Lu and Qian (2024), but departs in several important respects. They study the impact of the 2014 hukou reform on urban innovation using a difference-in-differences framework. They classify cities into treatment and control groups based on national policy guidelines: cities with populations exceeding five million were designated as control, while cities with smaller populations were grouped into different levels of treatment intensity. See Figure 1 to see the treatment groups. Their methodology assigns treatment status based purely on the city's total urban population after 2014.

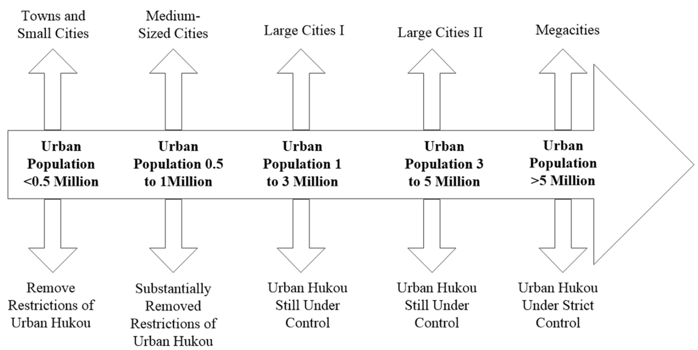

Figure 1: Treatment groups as described in Lu and Qian (2024).

However, this approach may misclassify cities due to the complexity of hukou reform implementation across different jurisdictions. Hukou policy changes in China are often determined at the county or district level within larger prefecture-level cities. For example, while Wenzhou was broadly categorized as a city required to strictly control population growth, in practice, several of its counties, such as Pingyang, actively promoted hukou liberalization during the reform period. Local press reports documented early cases of farmers successfully converting to urban hukou in these areas. As a result, a simple city-level population threshold may not accurately reflect the actual intensity of reform experienced by residents.

Moreover, the baseline classification of cities in Lu and Qian (2024) relies on the categorization of 282 cities, but it remains unclear how these cities were selected. Given that China has 707 cities at various administrative levels, it is likely that the authors excluded a substantial number of county-level and some prefecture-level cities. This selection issue raises concerns about the external validity of their findings.

In addition, Lu and Qian (2024) define the post-treatment period narrowly as 2015 and 2016. While a short post-reform window helps isolate immediate effects, hukou reform is a gradual process. Migrants' decisions to change household registration are influenced by social networks, property ownership, and family considerations, factors that typically adjust over multiple years. Local governments also implemented hukou reform at different times and speeds. Restricting the treatment period to only two years may understate the full adjustment dynamics following the reform.

In contrast to Lu and Qian (2024), this paper proposes a more detailed approach to classifying treatment exposure. Rather than relying solely on city population thresholds, I plan to incorporate case-by-case analysis of regional hukou policies where possible, focusing on representative provinces and cities. While full coverage of all 707 cities may not be feasible within the scope of this project, closer attention to local implementation will provide a more accurate measure of reform intensity. This approach aims to capture the heterogeneity of hukou reform and better assess its impact on urban innovation and population growth patterns.

3 Data and Methodology

3.1 Data Collection

Because the China Statistical Yearbook does not consistently publish city-level data, I manually collected data from the provincial statistical yearbooks released by individual provincial statistics bureaus. Each province manages its own website, data formatting, and access policies, requiring a province-by-province search to retrieve the relevant data. In some cases, access to provincial yearbooks was restricted for foreign IP addresses, and it was necessary to use a VPN connection through Hong Kong to successfully load the websites and download the documents.

For each city, I collected annual data on long-term resident population and GDP per capita from approximately 2010 to 2023, depending on data availability. Notably, the English translations provided in some statistical yearbooks were misleading. For example, in the Zhejiang Statistical Yearbook, the category labeled "total population" actually referred to the "registered resident population," rather than the actual number of residents living in the city. To measure the true size of the urban population, I used the "resident population" or "total population with permanent residence" indicators as the dependent variable.

3.2 Research Design

I estimate the effect of hukou reform on city-level population growth using a difference-in-differences (DiD) regression framework. The independent variable of interest is a treatment indicator for whether the city relaxed hukou restrictions following the 2014 national reform initiative. Treated cities are assigned a value of 1, and control cities, which either experienced no policy change or maintained strict hukou requirements, are assigned a value of 0. I define the post-treatment period as 2015 and subsequent years, assigning a value of 1 to observations in 2015 or later, and 0 otherwise.

The dependent variable is the log of the long-term resident population. To control for other factors influencing population growth, I include GDP per capita as a covariate. The baseline regression specification is as follows:

$$\log(\text{Pop}_{ipt}) = \alpha + \beta \left(\text{Treated}_i \times \text{Post}_t\right) + \theta \log(\text{GDPpc}_{ipt}) + \lambda_p + \delta_t + \varepsilon_{ipt}, \tag{1}$$

where ResidentPop_{it} represents the long-term resident population in city i at year t, Treated_i is a dummy variable equal to 1 for treated cities, Post_t is a dummy variable equal to 1 for post-reform years, γ_i are city fixed effects, δ_t are year fixed effects, λ_p are province fixed effects, and ϵ_{it} is the error term. The coefficient of interest is β, which captures the average treatment effect of hukou relaxation on population growth.

3.3 Treatment Assignment

Given the heterogeneity in local implementation of hukou reform policies, I conducted a manual review of local Chinese news sources and government announcements to determine the appropriate treatment status for each city. While some cities explicitly published their interpretation of the State Council's "Guiding Opinions on Further Deepening the Reform of the Household Registration System," others issued local guidelines extending urban hukou access to rural migrants, removing urban-rural hukou distinctions, or allowing long-term residents to apply for hukou where they habitually reside. Based on this detailed review, I assigned treatment labels case-by-case, rather than relying solely on national population-based classification thresholds. This approach aims to capture more accurately the variation in hukou policy implementation across cities and over time.

4 Results

4.1 Qualitative city–level patterns

Zhejiang. Figure 2 shows that the two *sub-provincial* cities, Hangzhou and Ningbo, dominate Zhejiang's demographic landscape. Their elevated administrative rank grants them greater fiscal autonomy and direct approval authority for many investment projects, advantages that translate into richer public services and a persistent population pull. By contrast, the remaining nine *prefecture-level* jurisdictions—Wenzhou, Jiaxing, Huzhou, Shaoxing, Jinhua, Quzhou, Taizhou, Lishui, and Zhoushan—must route major spending decisions through the provincial government; this institutional difference helps explain their more modest (and in several cases stagnating) population trajectories.

A closer look at Wenzhou, Shaoxing, and Taizhou illustrates the mismatch between the State Council's headline guidance and ground-level practice. All three exceeded the five-million threshold in 2014 and were therefore officially placed in the "strict control" category. Yet contemporaneous local reports tell a different story. In Wenzhou's coastal county of Pingyang, newspapers enthusiastically announced that "agricultural household registration will disappear next week," celebrating the first farmers to secure urban hukou. Shaoxing's municipal government issued implementation opinions that (i) abolished the agricultural / non-agricultural divide and (ii) lowered residence, employment, and social-insurance hurdles, explicitly targeting long-term migrants with "strong employment ability." Taizhou went further: public-security data indicated that 4.85 million agricultural hukou holders—roughly 82% of the local population—would be converted to unified resident status. These cases confirm that hukou policy in Zhejiang is ultimately forged at the county and district levels, and that headline population caps mask substantial local experimentation.

Sichuan. Sichuan's settlement pattern is even more polarised (Figure 4). Chengdu—a sub-provincial city and the province's sole megacity—continued to post rapid resident-population gains throughout the sample, despite retaining comparatively tight welfare eligibility for non-local hukou holders. Its thick labour market, deep university cluster, and status as Western China's transport hub appear to outweigh entry-barrier considerations. In contrast, prefecture-level Nanchong and Dazhou both qualified for the "strict control" list in 2014, yet their shrinking populations prompted local governments to loosen settlement rules in an effort to stem out-migration. Taken together, the qualitative evidence reinforces the need to measure treatment at a finer geographic scale than the national guideline's population bands.

4.2 Quantitative evidence

Table 1: Summary Statistics of Cities Included in the Dataset

Variable	**Median**	**Min**	**Max**
GDP per Capita (RMB)	77,803.5	29,517	200,278
Long-Term Resident Population (10,000s)	344.6	82.5	2,487.45
Urban Residents (%)	58.23%	32.92%	100.00%
10-Year Population Growth Rate (%)	2.05%	−35.78%	36.82%
Hukou Population (%)	107.88%	60.96%	146.87%

Notes: Statistics from provincial statistical yearbooks from Beijing, Shanghai, Zhejiang, and Sichuan. Resident-population series use the long-term resident measure; see text for details.

Table 1 presents median summary statistics. Medians are preferable to means because Beijing and Shanghai—which remain in the dataset for comparability tests—skew the distribution of both GDP per capita and urbanisation rates. Even after the 2014 reform, roughly one in five cities reports ten-year *negative* population growth, underscoring that policy relaxation is not a panacea for smaller inland centres. The median hukou-to-resident ratio exceeds 100%, reflecting the still-prevalent phenomenon of residents holding their hukou elsewhere.

The city-by-city trajectories plotted in Figures 2–5 flesh out these aggregate patterns. Within Zhejiang, Hangzhou and Ningbo capture the lion's share of post-2014 inflows, whereas Wenzhou's resident count barely edges upward. Jinhua, despite its smaller initial base, records one of the fastest proportional increases. GDP per capita dips around 2016 for several manufacturing-heavy prefectures, mirroring the nationwide deceleration after the supply-side reform

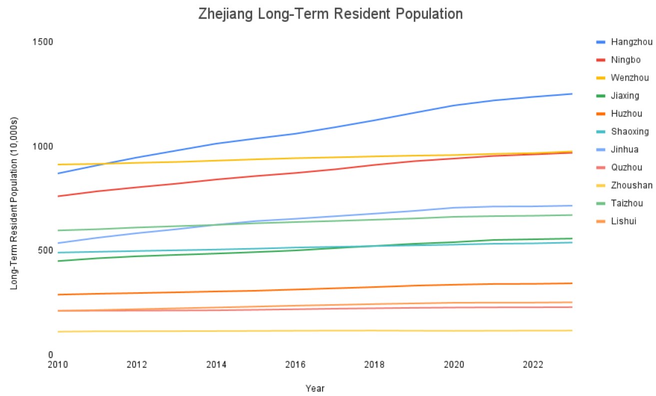

Figure 2: Permanent resident population for Zhejiang by city. Data is from the Zhejiang Statistical Yearbook.

campaign. In Sichuan, Chengdu's ascent contrasts sharply with the slow erosion of populations—and purchasing power—in most other prefecture seats.

Finally, Table 2 reports the benchmark difference-in-differences estimates. The interaction coefficient on *Treated* × *Post* is negative and marginally insignificant at the 10% level ($p = 0.069$). Point-wise, the estimate implies a 14.7% *lower* population level for treated cities relative to the counterfactual, once fixed effects and GDP controls are included. While the direction contradicts headline claims that the 2014 reform spurred migration into smaller cities, the confidence interval is wide ($[-30.4\%, 1.1\%]$), underscoring considerable uncertainty.

5 Discussion

The mixed quantitative findings mirror the institutional complexity uncovered in the qualitative review. China's multi-layered governance structure delegates crucial hukou decisions to counties and even township police stations, producing a mosaic of rules that no single population threshold can capture. In Zhejiang, county-level actors such as Pingyang embraced aggressive relaxation to counter labour shortages, whereas economically vibrant Wenzhou city itself opted for only incremental change. The Sichuan pattern is starker: Chengdu's agglomeration advantages dwarf the deterrent effect of its still-stringent welfare access rules, while neighbouring prefectures loosen restrictions simply to slow demographic decline. These asymmetries likely attenuate the measured average treatment effect and help explain the imprecise DiD coefficient.

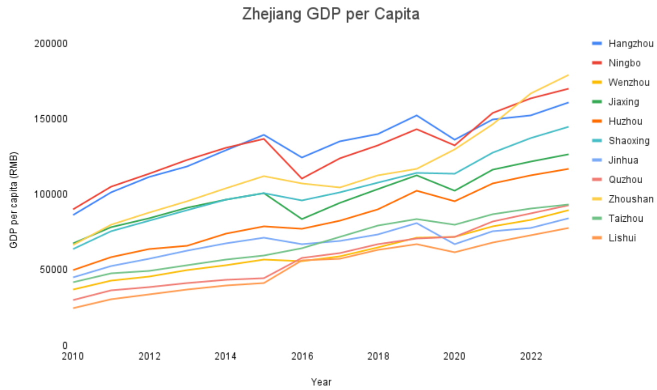

Figure 3: GDP per capita for Zhejiang by city. Data is from the Zhejiang Statistical Yearbook.

5.1 Policy interpretation

Taken together, the evidence suggests that the 2014 national hukou directive functioned more as a *permission slip* than as a binding mandate. Cities with strong fiscal positions (Hangzhou, Chengdu) could afford to remain selective, betting that migrants would come regardless. Peripheral counties facing population loss interpreted the same document as an invitation to open the gates. This heterogeneity implies that any one-size-fits-all evaluation may mislead central planners about the true elasticity of migration with respect to hukou policy.

5.2 Limitations and future research

The evidence assembled here is necessarily partial. Because the dataset spans only Zhejiang and Sichuan, it cannot speak to coastal super-cities such as Guangzhou or to inland regions where hukou incentives interact with poverty-alleviation programmes. Even within the two sampled provinces, reform intensity is coded as a simple binary, despite the fact that some jurisdictions merely re-labelled rural and urban hukou while others granted full access to public schooling and subsidised housing. Measurement error may also arise from inconsistent definitions of "resident population" across provincial yearbooks and from migrants who acquire an urban hukou yet continue to live elsewhere, thereby inflating origin-city counts. A broader study that incorporates all thirty-one provincial-level units, constructs an ordinal index of benefits granted, and links hukou changes to individual Census micro-data would address these concerns. Staggered adoption across cities could then be exploited in an event-study framework, and firm-level employment records could shed light on whether labour mobility is matched by capital reallocation.

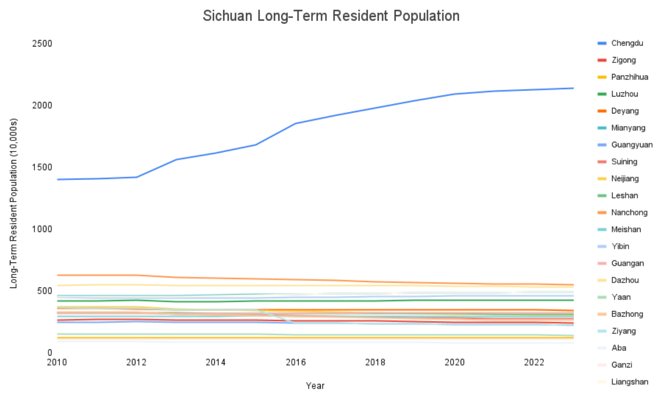

Figure 4: Permanent resident population for Sichuan by city. Data is from the Sichuan Statistical Yearbook.

6 Conclusion

This paper shows that the 2014 hukou reform, far from triggering a uniform population surge in smaller cities, produced a patchwork of outcomes dictated by local fiscal strength and agglomeration advantages. While sub-provincial centres such as Hangzhou and Chengdu continued to attract migrants even under tight settlement rules, several prefecture-level counties that fully abolished the rural–urban divide failed to reverse demographic decline. The average treatment effect—statistically indistinguishable from zero—thus masks pronounced heterogeneity. These findings imply that household-registration policy functions less as a centrally imposed lever and more as a discretionary tool whose efficacy hinges on local conditions.

Future work that expands geographic coverage, refines the measurement of reform depth, and links hukou changes to firm behaviour will be crucial for understanding how mobility policy can foster balanced regional development. Until such evidence is available, researchers should resist blanket categorizations of hukou policy status, and instead recognize the heterogeneity in China's national- to local-level administration.

References

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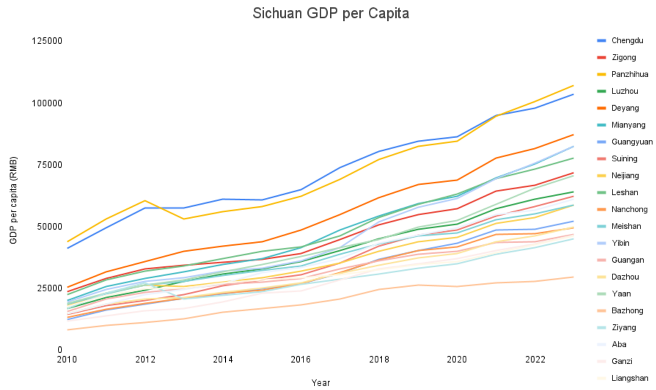

Figure 5: GDP per capita for Sichuan by city. Data is from the Sichuan Statistical Yearbook.

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Table 2: Difference-in-Differences Regression of log(population)

	Coef.	Std. Err.	z	p	[95% CI]
Treated	-0.9546	0.259	-3.69	0.000	$[-1.462, -0.448]$
Post	0.1019	0.194	0.52	0.600	$[-0.279, 0.483]$
Treated×Post	-0.1465	0.081	-1.82	0.069	$[-0.304, 0.011]$
log(GDP)	0.0426	0.319	0.13	0.894	$[-0.583, 0.669]$
Fixed effects					
Province FE	Yes				
Year FE	Yes				
Observations					476
R^2					0.569
Adj. R^2					0.551
F–statistic					5.15×10^{10}
Prob $> F$					<0.001
Cluster-robust SEs					Yes

Notes: Dependent variable is log(population). Standard errors are clustered at the appropriate level and reported in parentheses.